DIGITAL PIONEERS OF EGYPT INITIATIVE - PHASE 5
Data Scientist Track | Graduation Project

CUSTOMER CHURN INTELLIGENCE & RETENTION SYSTEM

From reactive firefighting to evidence-based customer retention

Project Documentation
Part 1: Project Planning & Management
Part 2: Literature Review
Part 3: Requirements Gathering

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1. Project Planning & Management

1.1 Project Overview

Customer churn is the event in which a customer stops using or paying for a company's service. In subscription businesses (telecom, internet providers, SaaS, streaming, gyms, insurance, online education) revenue is *recurring*: every customer who leaves removes future monthly revenue, and the company must spend marketing and sales budget to replace them.

The Customer Churn Intelligence & Retention System is an end-to-end analytics solution. It organizes available customer, subscription and billing data, queries it with SQL, cleans it, explores observed churn patterns, compares classification models, and presents results in an interactive dashboard. Risk scores should be described as churn propensity within the available data; forecasting future churn requires timestamped history and a defined prediction horizon. Support data will be used only if it is available.

1.2 The Real-World Business Problem

Almost every subscription company faces the same operational pain. The problem is not simply that customers leave; it is that the company **finds out too late and acts blindly**. The table below describes the concrete symptoms retention and customer-success teams report in practice, and how this project addresses each one.

#	Pain point inside companies	Why it hurts	How this project addresses it
1	Churn is discovered only **after** the cancellation or the missed payment.	The customer has already decided; win-back is far harder than prevention.	If historical data supports it, score active customers for a defined future period.
2	Retention offers (discounts, free upgrades, calls) are sent to everyone or to random customers.	Budget is wasted on loyal customers who would have stayed anyway, while real at-risk customers are missed.	A ranked risk list lets the team spend the limited retention budget on the highest-risk segment first.
3	Customer data is scattered across billing, CRM and support systems.	Nobody sees the full picture of a customer; analysis takes days of manual Excel work.	A relational database and reusable SQL queries join all sources into one analysis view.
4	Managers rely on intuition ("customers on monthly contracts probably leave more").	Decisions are not evidence-based and cannot be measured or defended.	EDA quantifies churn rate by contract, tenure, charges, services and support activity.
5	Reports are static spreadsheets that are outdated the moment they are sent.	Teams cannot filter by segment or drill down to individual customers.	An interactive dashboard with KPIs, filters and a downloadable at-risk customer list.
6	Existing predictive models are black boxes judged by accuracy only.	With ~25% churners, a model that always predicts 'no churn' is ~75% accurate but useless.	Models are compared with precision, recall, F1, ROC-AUC, confusion matrix and lift, with a justified threshold.

Why this problem matters financially

Retention economics have been studied for decades. Reichheld and Sasser (1990) argued that small improvements in customer retention can produce disproportionately large profit improvements, and later business writing (for example Gallo, Harvard Business Review, 2014) commonly states that acquiring a new customer costs several times more than keeping an existing one. The exact multiplier differs by industry, so this project does **not** rely on any single figure; instead it provides the analysis that lets a company compute its own numbers.

Illustrative example (hypothetical numbers for explanation only, not real company data):

Quantity	Value	Calculation
Active subscribers	10,000	Given
Average revenue per customer per month	$50	Given
Churn rate in the observed period	26%	Given (similar to the public IBM Telco sample)
Customers lost	2,600	10,000 x 0.26
Monthly recurring revenue lost	$130,000	2,600 x $50
Annualized revenue lost	$1,560,000	$130,000 x 12
If the model helps save only 10% of churners	260 customers = $156,000 per year	2,600 x 0.10 x $50 x 12

Key insight: The value of the system is not in predicting churn perfectly. It is in helping a team with a **limited retention budget and limited call-center capacity** contact the customers where intervention has the highest expected return.

1.3 Problem Statement

A subscription-based company is losing a significant share of its customers every period. The retention team does not know (a) **which customers** are most likely to leave, (b) **which characteristics** (contract type, tenure, price, services, support experience) are associated with leaving, or (c) **where to invest** its limited retention budget. Data exists in several tables, but it is not cleaned, not combined and not presented in a form that business users can act on. As a result, retention is reactive, expensive and unmeasured.

1.4 Proposed Solution

The team will deliver a working educational prototype with five connected layers:

Layer	What it does	Output
1. Data & SQL layer	Organizes data into related tables and answers business questions with SQL.	Schema, SQL scripts, validated query results
2. Data preparation layer	Detects and fixes missing, duplicate, inconsistent and wrongly typed values using documented rules.	Clean analysis dataset + cleaning log
3. Analytics (EDA) layer	Calculates churn rate and compares churn across segments and variables.	Charts, tables, insights
4. Prediction layer	Trains and compares a baseline and alternative classifiers; produces a churn-risk score per customer.	Model comparison table, metrics, scored customers file
5. Presentation layer	Interactive dashboard with KPIs, patterns, model metrics, risk ranking, filters and CSV export.	Dashboard + README + recommendations

1.5 Project Objectives (SMART)

ID	Objective	Measure of completion
O1	Build or adopt a relational customer database and query it with SQL.	At least 10 business questions answered with SQL; outputs validated against expected row counts.
O2	Clean missing, duplicated, inconsistent and incorrectly typed values.	Every data issue recorded in a cleaning log; row counts before/after reported; no silent record loss.
O3	Perform EDA to understand churn behavior.	Overall churn rate + at least 5 meaningful churn comparisons with charts and written interpretation.

ID	Objective	Measure of completion
O4	Analyze relationships between churn and tenure, contract, charges, services and support data (when available).	Statistical summaries and correlation/association analysis documented; no causal claims.
O5	Train and compare classification models.	At least 2 models (baseline + alternative) evaluated on the same held-out test set.
O6	Evaluate with more than accuracy.	Precision, recall, F1, ROC-AUC, confusion matrix and lift reported; threshold justified.
O7	Create an interactive dashboard with customer-level risk predictions.	Reviewer can open it, filter segments and see a ranked at-risk list without running notebooks.
O8	Document assumptions, limitations and retention recommendations.	Final report contains a limitations section and at least 3 evidence-based recommendations.

1.6 Project Scope

In scope

- Dataset selection, field documentation, data-quality checks, cleaning and EDA.
- Relational database design (or use of an existing relational database) and SQL analysis.
- A baseline classifier (Logistic Regression) and at least one alternative (Decision Tree and/or Random Forest; optionally a gradient-boosting model).
- Evaluation with precision, recall, F1-score, ROC-AUC, confusion matrix, and a decile-lift analysis.
- Cost-aware threshold discussion: explaining the business cost of false positives versus false negatives.
- Interactive dashboard: summary cards, churn charts, model section, filterable customer risk table, definitions and caveats.
- Reproducible repository: code, requirements file, README, documentation PDF, data-source notes.

Out of scope

- Automatically contacting customers or changing subscriptions or prices.
- Production deployment connected to a real company's live customer database.
- Claiming that a variable *causes* churn from observational data alone.
- Guaranteeing that every high-risk prediction will be correct.
- Real-time streaming prediction, A/B testing infrastructure, or measuring actual retention campaign results.

1.7 Assumptions and Constraints

Type	Statement
Assumption	A public or synthetic educational dataset (for example the IBM Telco Customer Churn sample) will be used; no real personal data is required.
Assumption	The churn column represents customers who left during the observed period; the definition will be documented exactly.
Assumption	If the chosen dataset is a single flat file, the team will normalize it into relational tables (Appendix A) so that SQL joins are meaningful. Where a support or transaction table does not exist in the data, this limitation will be stated explicitly.
Constraint	Fixed deadline for the documentation; the documentation PDF must be uploaded to the team GitHub repository by **Monday 19-10-2026**, then the implementation continues according to the work plan.

Type	Statement
Constraint	Five team members with different skill levels; workload must be balanced and every member must understand the full pipeline.
Constraint	Standard student laptops; no paid cloud services required.

1.8 Proposed Tools and Technologies

Area	Tool	Purpose / justification
Database / SQL	SQLite, MySQL or PostgreSQL	Store related tables; SQLite requires no server and is easiest for reproducibility.
Data processing	Python, pandas, NumPy	Cleaning, transformation, feature engineering.
Visualization / EDA	Matplotlib, Seaborn, Plotly	Static and interactive charts.
Machine learning	scikit-learn (optionally XGBoost / LightGBM)	Pipelines, models, metrics, cross-validation.
Imbalance handling	class_weight, stratified splits; SMOTE only inside training folds if justified	Avoid models that ignore the minority class.
Dashboard	Streamlit (primary) or Power BI	Interactive business dashboard with filters and export.
Notebook environment	Jupyter / Google Colab	Reproducible analysis notebooks.
Collaboration	Git and GitHub (branches, pull requests, issues, project board)	Version control and code review.
Documentation	Markdown + PDF	README, data dictionary, final report.

1.9 Methodology

The team will follow a simplified **CRISP-DM** lifecycle (Cross-Industry Standard Process for Data Mining), because it matches the project stages and is widely used in data science teams:

1. **Business understanding** - define the churn problem, goals and success criteria (this document).
2. **Data understanding** - inspect the dataset, define the target, check quality.
3. **Data preparation** - SQL extraction, cleaning, feature engineering, train/test split.
4. **Modeling** - train baseline and alternative classifiers inside leakage-safe pipelines.
5. **Evaluation** - compare models on the same held-out test set with multiple metrics; discuss business cost.
6. **Deployment (prototype level)** - dashboard, README and final report. No production deployment.

Work is organized in weekly iterations. Progress is tracked on a GitHub Projects board with columns *Backlog, In Progress, In Review, Done*. Each task is a GitHub issue with an owner and a due date.

1.10 Work Plan and Timeline

The schedule below is a proposed duration plan and must be adjusted to the official course deadlines. The documentation sprint (Week 0) is separate from the implementation, which is planned over **five weeks (35 days)**. The core technical work needs at least 20 working days; the remaining time is deliberate **buffer** for debugging, integration problems and review. Tasks marked 'parallel' can start at the same time, but model evaluation must wait for finished data cleaning.

Phase	Duration	Main activities	Deliverables / milestone
Week 0 Documentation sprint	3 days (deadline: 19-10-2026)	Choose project name; create the single team GitHub repository; register name and link in the Drive sheet; write and merge the three documentation parts; upload the PDF to GitHub; the Team Leader submits on the LMS.	**M0:** Documentation PDF in GitHub before the 19-10-2026 deadline.
Week 1 Setup and data	7 days	Confirm requirements; select dataset and document its fields; design the database schema; create repository structure; set up task board; review related work; agree on file formats between members.	Project plan, data dictionary, schema diagram, initial repository. **M1:** data loaded into the database.
Week 2 SQL and cleaning	7 days	Write and validate SQL queries (Appendix B); profile data; define and apply cleaning rules; document the cleaning log.	SQL scripts, cleaning log, first clean dataset. **M2:** clean dataset frozen (version-tagged).
Week 3 EDA and features	7 days	Complete EDA charts and written interpretation; feature engineering; stratified split; baseline Logistic Regression running inside a pipeline.	EDA notebook and charts, feature pipeline, baseline results. **M3:** baseline model evaluated.
Week 4 Modeling and prototype	7 days	Train alternative models; cross-validation; threshold and lift analysis; error analysis; export scored customers; build the first dashboard prototype to test integration.	Model notebook, comparison table, scored customers CSV, prototype dashboard. **M4:** first end-to-end run.
Week 5 Finalization and buffer	7 days	Finish dashboard (KPIs, filters, export); write recommendations and limitations; clean-environment test; finalize README, report and presentation; use remaining days as buffer for fixes.	Working prototype, final report PDF, README, presentation. **M5:** final delivery.

Detailed task breakdown by week

Week	Task	Owner	Effort
0	Write Project Planning & Management; merge all sections	Member 1	1 day
0	Write Literature Review (sources, comparison, gap)	Members 2 and 3	1 day
0	Write Requirements Gathering (functional, non-functional, data, model)	Members 4 and 5	1 day
1	Dataset selection and data dictionary	Members 2, 3	3 days
1	Database schema, keys, relationships, load script	Member 2	4 days
1	Repository structure, README skeleton, branching rules, task board	Members 1, 5	3 days
2	SQL business-question scripts and validation	Member 2	5 days
2	Data profiling, cleaning rules and cleaning log	Member 3	5 days
3	EDA charts and written interpretation	Members 3, 2	5 days
3	Feature pipeline, train/test split, baseline model	Member 4	5 days
3	Dashboard layout design (wireframe) and chosen tool setup	Member 5	4 days
4	Alternative models, cross-validation, threshold, lift, error analysis	Member 4	5 days
4	Prototype dashboard connected to sample outputs	Member 5	5 days
4	Draft recommendations and limitations	Members 1, 3	3 days
5	Final dashboard: KPIs, filters, risk table, export	Member 5	4 days
5	Final report, README, presentation	Members 1, 3	4 days
5	End-to-end test on a clean environment; presentation rehearsal	All	2 days
5	Buffer for fixes and review	All	2-3 days

1.11 Team Structure and Task Allocation

Each member owns one primary workstream, but **every member must be able to explain the whole pipeline** and must review at least one pull request from another member. Roles can be swapped according to team strengths.

Member	Primary responsibility	Specific tasks and outputs
Member 1 - Project Lead & Documentation	Coordination, planning, documentation	Maintain timeline and task board; organize weekly meetings; write Project Planning & Management; merge document sections; track deadlines; ensure consistency of terminology; submit on the LMS as Team Leader.
Member 2 - SQL & Database	Database design and SQL analysis	Design tables, primary/foreign keys and relationships; load data; write and validate queries for churn rate, segments, contracts, billing and support patterns; document outputs and expected counts.
Member 3 - Data Cleaning & EDA	Data quality and exploration	Profile missing values, duplicates, types and inconsistent categories; build the cleaning log and clean dataset; create EDA charts; summarize churn relationships in plain language.
Member 4 - Machine Learning	Prediction and evaluation	Define target; build a leakage-safe preprocessing pipeline; train and tune baseline and alternative models; export predictions and scores.
Member 5 - Dashboard & Integration	Dashboard and repository integration	Build dashboard with KPIs, charts and risk ranking; integrate team outputs; maintain README and requirements; run final environment test. With Member 4, check model metrics, threshold trade-offs and false positives/negatives.

RACI matrix (R = Responsible, A = Accountable, C = Consulted, I = Informed)

Activity	M1	M2	M3	M4	M5
Project plan and timeline	A/R	C	C	C	C
Database schema and SQL queries	I	A/R	C	C	I
Data cleaning and cleaning log	I	C	A/R	C	I
EDA and insights	I	C	A/R	C	C
Feature engineering and modeling	I	C	C	A/R	C
Model evaluation and threshold	I	I	C	A/R	R
Dashboard	I	C	C	C	A/R
Final documentation and report	A/R	R	R	R	R
Repository, README, final testing	C	C	C	C	A/R

1.12 Communication and Collaboration Workflow

- **Meetings:** one 30-45 minute weekly planning meeting and two short progress check-ins per week (online).
- **Channel:** one team group chat for quick questions; decisions are recorded as GitHub issues so they are not lost.
- **Git workflow:** the *main* branch is always runnable; each member works on a feature branch (for example *feature/sql-queries*); changes enter main only through a pull request reviewed by at least one other member.
- **Commit rules:** small, frequent commits with meaningful messages; never commit passwords, API keys or restricted data.

- **Definition of Done:** a task is done when its output is committed, documented, reviewed, and reproducible by another member.

1.13 Risks and Mitigation

ID	Risk	Prob.	Impact	Mitigation	Contingency
R1	Missing, duplicated or inconsistent data	High	High	Profile data in Weeks 1-2; documented cleaning rules; validate row counts and key fields.	Fall back to simple documented imputation; report affected rows.
R2	Class imbalance (fewer churners)	High	High	Stratified split; inspect distribution; use recall, precision, F1, ROC-AUC; class weights.	Compare threshold choices; report PR curve.
R3	Data leakage inflates results	Medium	High	Split before preprocessing; use scikit-learn Pipelines; fit transformers on training data only; remove fields that reveal the outcome.	Re-run evaluation after fixing the pipeline.
R4	Uneven workload or limited time	Medium	High	Owners and intermediate deadlines; weekly review; GitHub issues board.	Re-assign tasks; reduce optional features (FR-14).
R5	Dashboard and model do not integrate	Medium	High	Agree on file formats (CSV/Parquet + joblib model) in Week 1; first integration by Week 4.	Dashboard reads a pre-scored CSV instead of loading the model.
R6	Privacy misuse of customer data	Low	High	Use public/synthetic/anonymized data; keep identifiers anonymized; no credentials in GitHub.	Remove sensitive files from history and rotate secrets.
R7	Dataset lacks support/transaction tables	High	Medium	State limitation clearly; derive proxy tables (for example billing summary) without pretending they are real transactions.	Add a second public dataset only if time allows.
R8	Merge conflicts and broken main branch	Medium	Medium	Pull requests, small commits, branch protection.	Revert to last tagged stable version.
R9	Team member unavailable	Medium	Medium	Pair on critical tasks; each member documents their work.	Backup owner takes over (see RACI).
R10	Over-claiming model performance	Medium	Medium	No accuracy promises before experiments; report limitations honestly.	Rewrite claims to match measured results.

1.14 Success Criteria and KPIs

Process and quality KPIs

KPI	Target	How it is measured
Data quality	100% of identified data issues are logged and handled by reproducible rules	Cleaning log reviewed; row counts before/after reconcile
SQL coverage	>= 10 business questions answered; every query output checked	Appendix B checklist; expected-count tests
EDA coverage	Churn rate + >= 5 meaningful churn comparisons with interpretation	EDA notebook and dashboard charts
Model comparison	>= 2 approaches (including a baseline) on the same held-out test set	Comparison table in the notebook and dashboard
Model reporting	Precision, recall, F1, ROC-AUC and confusion matrix reported; not accuracy alone	Evaluation section of the report

KPI	Target	How it is measured
Dashboard usability	A first-time reviewer can find overall churn rate and the top-10 at-risk customers in under 2 minutes without reading code	Usability test with 2-3 classmates; timings recorded
Reproducibility	A teammate can run the project from a clean clone by following the README only	Clean-environment test in Week 5
Team delivery	All three documentation parts complete; PDF in the GitHub repository by 19-10-2026	Repository check and LMS submission

Business-oriented model KPIs (reported, not promised in advance)

- **Improvement over baselines:** the best model should perform better than a majority-class guess and be compared with the Logistic Regression baseline on the same test set.
- **Lift / capture rate:** the share of all churners found within the top 10% and top 20% highest-risk customers, compared with random selection.
- **Recall for churners** at the chosen threshold, together with the precision at that threshold (so the cost of false alarms is visible).

Important: final numeric model targets (for example a minimum ROC-AUC) will be set only after inspecting the dataset and the baseline results. The team will not promise a specific accuracy before experiments are run.

1.15 Project Deliverables

Deliverable	Format	Owner
Project documentation (3 parts)	PDF in GitHub /docs	Member 1
Data dictionary and dataset notes	Markdown in /data	Members 2, 3
Database schema + load script	SQL in /sql	Member 2
SQL business-question scripts	SQL in /sql	Member 2
Cleaning log + clean dataset	Markdown / CSV	Member 3
EDA notebook and charts	Jupyter notebook	Member 3
Model notebook, comparison table, scored customers	Notebook + CSV + saved model	Member 4
Interactive dashboard	Streamlit app (or Power BI file)	Member 5
README and requirements	Markdown + requirements.txt	Member 5
Final report and presentation	PDF and slides	All

2. Literature Review

2.1 Purpose and Search Approach

This review looks for projects, tools, tutorials and research that address customer churn, in order to learn what already works, where it falls short, and how this project can combine the best ideas. The review has three guiding questions:

1. **Representation:** how is churn data structured and how is the churn label defined?
2. **Modeling:** which classification approaches are used and how are they prepared?
3. **Evaluation and action:** how is model quality measured, and how are predictions turned into retention decisions?

Sources were chosen from three groups: (1) public educational implementations, (2) technical documentation of standard tools, and (3) peer-reviewed or widely cited academic and business papers. This is a focused practical review for a capstone project, not a systematic academic literature review.

2.2 Related Work

A. IBM Telco Customer Churn sample dataset and Cognos dashboard

IBM publishes a sample dataset for a fictional telecommunications company, together with an analytics dashboard example. The data describes customer profile, subscribed services, contract and billing information, and a churn field showing whether the customer left. The widely used public version has roughly seven thousand customers and twenty-one columns, with about one quarter of customers marked as churned.

- **Strengths:** clearly structured, free, well known, suitable for segmentation and dashboarding practice.
- **Limitations:** fictional telecom context; a single static snapshot, so it cannot support true time-based forecasting; no real support-ticket or transaction history; observational data cannot prove causation.
- **What we adopt:** analysis of churn by contract, tenure, charges and services; presentation of findings in a dashboard.
- **How we differ:** we add a relational database with SQL analysis, a documented cleaning stage, model comparison, customer-level risk ranking and a reproducible repository.

Sources: IBM, 'Telco customer churn' [1]; IBM sample dataset repository [4].

B. IBM tutorial: Predict telecommunications churn

The tutorial frames churn as a binary classification problem and walks through inspecting data, preparing the target and input fields, building a model and producing predictions.

- **Strengths:** a clear workflow from data to scored records.
- **Limitations:** a tutorial does not by itself demonstrate generalization, fairness or business impact; evaluation depth is limited.
- **What we adopt:** churn as a binary target; train/test separation; customer-level probabilities.
- **How we differ:** multiple models compared on one test set, leakage-safe pipelines, imbalance handling, threshold and cost discussion.

Source: IBM, 'Predict telecommunications churn' [2].

C. scikit-learn evaluation and modeling documentation

The scikit-learn documentation defines precision, recall, F1-score, ROC-AUC and the confusion matrix and provides tools (pipelines, stratified splitting, cross-validation) for transparent comparison. For churn, **recall** shows how many actual churners were found, **precision** shows how many flagged customers really churned, **F1** balances the two, and **ROC-AUC** measures how well the model ranks churners above non-churners across thresholds.

- **Strengths:** standard, reproducible definitions.
- **Limitations:** no single metric captures business cost; the threshold must be chosen using the company's retention capacity and the cost of false alarms versus missed churners.
- **What we adopt:** multi-metric evaluation, confusion matrix, stratified split, pipelines.

Source: scikit-learn, 'Metrics and scoring' [3].

D. Academic research on churn prediction (Verbeke et al., 2012)

Verbeke, Dejaeger, Martens, Hur and Baesens compared many classification techniques for telecom churn and emphasized **profit-driven** evaluation: the usefulness of a model depends on campaign cost and the value of the retained customer, not only on a statistical score.

- **Strengths:** rigorous comparison across techniques; links modeling to business value.
- **Limitations:** uses proprietary operator data and sophisticated cost assumptions that a student team cannot replicate exactly.
- **What we adopt:** the idea that threshold selection must reflect cost; we present a simple expected-value threshold analysis with clearly labeled assumptions.

Source: Verbeke et al. [5].

E. Ensemble tree models: Random Forest and gradient boosting

Breiman (2001) introduced Random Forests, which average many decision trees and often perform well on tabular data with mixed variable types. Chen and Guestrin (2016) introduced XGBoost, a scalable gradient-boosting system that is widely used for tabular prediction tasks.

- **Strengths:** capture non-linear relationships and interactions; provide feature-importance information.
- **Limitations:** less interpretable than Logistic Regression; can overfit without validation; feature importance can be misleading with correlated variables.
- **What we adopt:** Logistic Regression as an interpretable baseline plus a tree-based alternative; cross-validated comparison.

Sources: Breiman [6]; Chen and Guestrin [7].

F. Handling class imbalance and metric choice

Churn datasets usually contain fewer churners than non-churners. Chawla et al. (2002) proposed SMOTE, which creates synthetic minority-class examples; Saito and Rehmsmeier (2015) showed that precision-recall plots can be more informative than ROC plots on imbalanced data.

- **What we adopt:** stratified splitting and class weighting first; SMOTE only if justified and only inside training folds to avoid leakage; report a precision-recall view in addition to ROC-AUC.

Sources: Chawla et al. [8]; Saito and Rehmsmeier [9].

G. Commercial customer-success and retention platforms

Commercial customer-success platforms (for example Gainsight and ChurnZero) and CRM-integrated analytics offer health scores, alerts and workflows for retention teams. They show the real market need for

early-warning and prioritization.

- **Strengths:** integrated with company systems; automated alerts and workflows; support and usage data included.
- **Limitations:** paid, closed systems; not suitable for learning or for transparent, explainable academic work; scores are often hard to inspect.
- **What we adopt:** the idea of a customer risk list and segment-level indicators for non-technical users.
- **How we differ:** our project is open, transparent and reproducible, and it documents every cleaning, modeling and evaluation decision.

Note: platform descriptions are general; the team will verify current product details from the vendors' public websites before the final report.

H. Retention economics

Reichheld and Sasser (1990) argued that retaining customers has a strong effect on profitability, and Gallo (2014) discussed why keeping the *right* customers matters. These sources motivate prioritization of retention effort rather than treating all customers equally.

Sources: Reichheld and Sasser [10]; Gallo [11].

2.3 Comparison of Related Work and the Proposed Project

Dimension	IBM sample / dashboard	IBM modeling tutorial	Commercial CS platforms	Proposed system
Main focus	Explore and visualize churn	Build a prediction workflow	Monitor customer health and trigger actions	End-to-end analytics + prediction + retention prioritization
SQL / database	Structured data, not the focus	Not the focus	Internal, hidden	Explicit relational schema and documented SQL queries
Cleaning and EDA	Starting point only	Brief preparation	Automated, not transparent	Dedicated stage with cleaning log and interpretation
Model comparison	Not covered	Single workflow	Proprietary scoring	Baseline vs alternative(s) on the same test set
Evaluation depth	Descriptive indicators	Basic	Not exposed	Precision, recall, F1, ROC-AUC, confusion matrix, lift, threshold analysis
Imbalance and leakage	Not discussed	Limited	Unknown	Stratified split, pipelines, class weights, documented checks
Customer-level output	Aggregated charts	Predictions	Health scores and alerts	Ranked risk list with filters and CSV export
Transparency / reproducibility	Medium	Medium	Low (closed)	High (open repository, README, fixed seeds)
Cost	Free	Free	Paid	Free (open-source stack)

2.4 Lessons Learned and Design Decisions

Lesson from the literature	Design decision in our project
Accuracy alone is misleading for imbalanced churn data.	Report precision, recall, F1, ROC-AUC, confusion matrix and lift; never judge by accuracy only.
Preprocessing before splitting leaks test information.	Split first; fit imputers, encoders and scalers on training data only through scikit-learn Pipelines.

Lesson from the literature	Design decision in our project
Business cost determines the useful threshold.	Provide a threshold slider and a simple expected-value analysis with labeled assumptions.
Interpretable models help adoption.	Use Logistic Regression as baseline; show coefficients or feature importance with the caveat that association is not causation.
Single static snapshots limit forecasting.	State the limitation; avoid claims about future timing of churn.
Retention budget is limited.	Rank customers by risk and show how many churners are found in the top 10% / 20%.

2.5 Research Gap and Project Contribution

The reviewed material demonstrates useful parts of churn analytics, but educational examples often stop at a dashboard or at a single model, while commercial tools hide their methods. The contribution of this project is **integration and transparency**: SQL-based business questions, documented data cleaning, EDA, comparative classification with proper evaluation, customer-level risk estimates and an interactive dashboard, all reproducible from one repository. The project does **not** claim a new machine-learning algorithm.

2.6 References

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Team note: the team should open each source once, confirm the citation details and add access dates before final submission.

3. Requirements Gathering

3.1 Stakeholders and Their Needs

Stakeholder	Role in the scenario	Needs and expectations
Retention / marketing manager	Primary business user	Know the current churn rate; identify high-risk segments; get a prioritized list of customers to contact within a limited budget; understand risk in plain language.
Customer-support lead	Business user	See whether support interactions relate to churn; identify customers with repeated issues (if support data exists).
Finance / revenue manager	Business user	Estimate revenue at risk; compare retention spending with expected savings.
Data analyst	Technical user	Access clean structured data; run SQL; inspect distributions; export charts and summaries.
Data scientist (team)	Technical user	Documented target; reproducible features; comparable models; error inspection.
Dashboard user	End user	View indicators, filter groups, interpret risk categories and see which factors matter.
Instructor / evaluator	Reviewer	Review methodology; reproduce the workflow; see assumptions, sources and individual contributions.
Data owner / privacy officer (indirect)	Governance	Assurance that personal data is protected and predictions are not misused.

3.2 User Stories

ID	User story	Linked requirement
US-01	As a retention manager, I want to see the overall churn rate and number of customers so that I understand the size of the problem.	FR-06, FR-11
US-02	As a retention manager, I want to see churn by contract type, tenure band and monthly charges so that I know which segments to focus on.	FR-06, FR-11, FR-13
US-03	As a retention manager, I want a ranked list of active customers by churn risk so that I contact the most at-risk customers first.	FR-10, FR-11
US-04	As a retention manager, I want to change the risk threshold and see how many customers are flagged so that I can match my team's capacity.	FR-15
US-05	As a finance manager, I want an estimate of monthly revenue held by high-risk customers so that I can judge the value of retention actions.	FR-16
US-06	As a data analyst, I want to run documented SQL queries so that I can validate numbers shown in the dashboard.	FR-03
US-07	As a data scientist, I want a leakage-safe pipeline and a model comparison table so that results are trustworthy.	FR-07, FR-08, FR-09
US-08	As a dashboard user, I want definitions and caveats so that I do not misread a risk score as a certainty.	FR-17, NFR-08
US-09	As an instructor, I want a README and requirements file so that I can reproduce the project.	FR-12, NFR-05
US-10	As a retention manager, I want to export the filtered at-risk list to CSV so that I can share it with the call team.	FR-14

3.3 Functional Requirements

Priority uses MoSCoW: **Must** (required), **Should** (important), **Could** (optional if time allows).

ID	Requirement	Priority	Verification
FR-01	The system shall load a structured customer dataset from an agreed source into the project database.	Must	Row count in database equals source row count.
FR-02	The team shall document dataset fields, data types, target variable, source, license and limitations in a data dictionary.	Must	Data dictionary file exists and covers all columns.
FR-03	The SQL component shall support queries across customer, subscription, billing/transaction and support information when those tables exist.	Must	Queries run without error; outputs checked against expected counts.
FR-04	The data preparation component shall detect missing values, duplicates, inconsistent categories and incorrect data types.	Must	Data profiling report produced.
FR-05	The system shall create a clean analysis dataset using documented, repeatable rules and a cleaning log with row counts before and after.	Must	Re-running the script produces identical output.
FR-06	The EDA component shall calculate overall churn rate and compare churn across available segments and variables.	Must	At least 5 comparisons with charts and interpretation.
FR-07	The ML component shall define churn as a binary target and create a stratified train/test split with a fixed random seed.	Must	Class proportions similar in train and test.
FR-08	The team shall train a baseline model and at least one alternative classifier.	Must	Both models evaluated on the same test set.
FR-09	The system shall report precision, recall, F1-score, ROC-AUC and the confusion matrix.	Must	Metrics table present in notebook and dashboard.
FR-10	The prediction output shall provide a churn-risk score (probability) and a risk category (for example Low / Medium / High) for each eligible customer.	Must	Scored customers file contains ID, score and category.
FR-11	The dashboard shall show summary cards, churn patterns, model metrics and risk-ranked customers or segments.	Must	Reviewer checklist.
FR-12	The repository shall include a README with setup and run instructions, dependencies, data source and team contributions.	Must	Clean-clone test.
FR-13	The dashboard should allow filtering by contract, tenure, services and other available fields.	Should	Filters change charts and table.
FR-14	The dashboard could export the filtered risk list or summaries to CSV.	Could	Downloaded file opens correctly.
FR-15	The dashboard should let the user change the risk threshold and show the resulting number of flagged customers, precision and recall.	Should	Slider updates values.
FR-16	The dashboard should show the monthly charges of high-risk customers (revenue at risk).	Should	Value equals SQL check.
FR-17	The dashboard shall include definitions: churn label, observation period, risk threshold, data source and limitations.	Must	Definitions panel visible.
FR-18	The ML component should report a decile lift or cumulative gains view showing churners captured in the top-risk groups.	Should	Lift table or chart present.

3.4 Non-Functional Requirements

ID	Quality attribute	Requirement / acceptance statement
NFR-01	Usability	Labels, units and definitions must be understandable to a non-technical manager; a first-time user finds the churn rate and top-10 risk list within 2 minutes.
NFR-02	Performance	Data loading, analysis and dashboard filtering complete in a reasonable time on a standard student laptop; actual timings are recorded during testing (target: filter updates in a few seconds).

ID	Quality attribute	Requirement / acceptance statement
NFR-03	Reliability	Repeated runs on the same data and configuration give consistent cleaning results and metrics, subject to documented random seeds.
NFR-04	Maintainability	Code is split into logical files/notebooks with meaningful names and comments for non-obvious decisions; no duplicated logic between notebook and dashboard.
NFR-05	Reproducibility	Dependencies and run instructions documented; seeds fixed; data source and cleaning steps traceable; project runs on a clean environment.
NFR-06	Security and privacy	No real personal data, passwords, API keys or credentials in GitHub; use public, synthetic or anonymized data; dashboard shows anonymized IDs only.
NFR-07	Data integrity	Customer IDs checked for duplicates; joins validated by row counts; transformations never silently discard records.
NFR-08	Interpretability	Risk scores explained as model estimates, possibly uncalibrated; correlation never presented as causation.
NFR-09	Fairness and responsible use	Demographic attributes (such as gender, senior-citizen flag) are reviewed; their use is justified or excluded; predictions must not be used for consequential decisions without further validation and governance.
NFR-10	Portability	The project runs on Windows, macOS or Linux using only open-source tools and a requirements file.
NFR-11	Compatibility	Dashboard works in a current version of a common browser (Chrome, Edge or Firefox).
NFR-12	Testability	Key functions (cleaning, feature pipeline, scoring) have at least simple checks, such as asserting row counts and column types.

3.5 Data Requirements

- **Minimum fields:** unique customer identifier, customer/subscription attributes and a clearly defined churn outcome.
- **Useful optional fields:** contract type, tenure, monthly and total charges, services, payment method, support tickets, complaints, transaction history.
- **Candidate dataset:** the public IBM Telco Customer Churn sample (single flat file, about 7,000 rows and 21 columns). The team will confirm the exact file, its license and row count in Week 1.
- **Known data issues to check:** the TotalCharges column is stored as text and may contain blank values for customers with zero tenure; some categories repeat the same meaning in different words (for example 'No internet service' and 'No phone service' versus 'No'); verify duplicates and data types.
- **Target definition:** document exactly what 'churn' means (for example 'customer left within the last month') and the observation period; do not mix labels from different time windows.
- **Data suitability:** if support or transaction tables are missing, state this limitation explicitly instead of implying they were analyzed.
- **Train/test integrity:** keep test data separate from fitting, feature selection and threshold selection (use validation or cross-validation on training data for tuning).
- **Leakage check:** remove any field that is only known after the customer has left (for example a churn reason or churn score column) from the model input.

3.6 Model and Evaluation Requirements

The prediction target is binary: **1 = customer churned**, **0 = customer stayed**.

Item	Requirement
Baseline model	Logistic Regression (interpretable) with scaling and one-hot encoding inside a pipeline.
Alternative models	Decision Tree and/or Random Forest; optional gradient boosting (XGBoost or LightGBM) if time allows.
Split	Stratified train/test split (for example 80/20) with a fixed random seed; test set used only for final evaluation.
Preprocessing	Imputers, encoders and scalers fitted on training data only, through scikit-learn Pipeline / ColumnTransformer.
Imbalance	Inspect class distribution; use class weights first; SMOTE only inside training folds and only if justified.
Tuning	Cross-validation on the training set (stratified k-fold); document the parameter grid.
Metrics	Confusion matrix, precision, recall, F1-score, ROC-AUC, precision-recall curve; decile lift.
Threshold	State the chosen decision threshold and why; default 0.5 is not assumed to be optimal.
Business interpretation	Explain false positives (a retention offer sent to a customer who would have stayed - wasted cost) and false negatives (a leaving customer not flagged - lost revenue).
Calibration	Explain that probabilities are estimates and may be poorly calibrated unless calibration is tested.
Explainability	Show coefficients or feature importance, with a note that these show association, not causation.
Error analysis	Inspect which segments the model misclassifies most.

Simple expected-value threshold analysis (illustrative assumptions to be set by the team)

For a chosen threshold, the expected benefit of a campaign can be estimated as:

Net benefit = TP x success_rate x customer_value - (TP + FP) x offer_cost
TP = churners correctly flagged; FP = non-churners incorrectly flagged.
success_rate, customer_value and offer_cost are assumptions, not measured facts.

Show these values as editable assumptions; do not present them as measured facts.

3.7 Dashboard Requirements

Section	Content	Requirement IDs
Header / summary cards	Total customers; churned customers; churn rate; number of high-risk customers at the selected threshold; monthly revenue at risk.	FR-11, FR-16
Churn patterns	Churn by contract type, tenure band, monthly charges band, internet service / add-on services, payment method; each chart has a short takeaway sentence.	FR-06, NFR-01
Model section	Selected model name; comparison table of all models; confusion matrix; ROC and precision-recall curves; decile lift.	FR-09, FR-18
Risk threshold control	Slider that updates the number flagged, precision, recall and estimated net benefit.	FR-15
Customer / segment view	Filterable, sortable table: anonymized customer ID, risk score, risk category and key explanatory fields; CSV export.	FR-10, FR-13, FR-14
Definitions and caveats	Churn definition, observation period, risk threshold, data source, limitations, and a warning against causal interpretation.	FR-17, NFR-08

The dashboard must not display unnecessary personal identifiers or sensitive fields.

3.8 Acceptance Criteria

Area	Acceptance test	Evidence
Data	Source, target, field definitions and cleaning decisions documented; row counts before and after cleaning reported.	Data dictionary and cleaning log
SQL	Core queries execute successfully; outputs match expected counts or manually validated samples.	SQL files + validation notes
EDA	Churn-rate summary and at least five interpretable comparisons supported by charts or tables.	EDA notebook
Model	At least two approaches compared on the same held-out test data with the selected metrics.	Comparison table
Threshold	Chosen threshold documented with its business justification.	Report section
Dashboard	A reviewer opens the dashboard, understands the key indicators and inspects patterns without running notebook cells.	Usability test notes
Repository	README, code, requirements, documentation PDF and data-source notes present; no secrets or unauthorized personal data committed.	Repository checklist
Reproducibility	A teammate reproduces results from a clean clone using only the README.	Test log

3.9 Traceability Matrix

Objective	Functional requirements	Non-functional requirements	Acceptance area
O1 Database and SQL	FR-01, FR-03	NFR-07	SQL
O2 Data cleaning	FR-02, FR-04, FR-05	NFR-03, NFR-07	Data
O3 EDA	FR-06	NFR-01	EDA
O4 Relationships with churn	FR-06	NFR-08	EDA
O5 Model comparison	FR-07, FR-08	NFR-03, NFR-05	Model
O6 Evaluation beyond accuracy	FR-09, FR-18	NFR-08	Model, Threshold
O7 Dashboard and predictions	FR-10, FR-11, FR-13 to FR-17	NFR-01, NFR-02, NFR-06, NFR-11	Dashboard
O8 Documentation and recommendations	FR-02, FR-12	NFR-04, NFR-05, NFR-09	Repository

Appendix A - Proposed Relational Database Schema

If the chosen dataset is one flat file, the team will split it into the following related tables so that SQL joins and referential integrity can be practiced. Tables that cannot be filled from real data (for example support tickets) will be marked as 'not available' in the documentation rather than invented.

Table	Key columns	Description
customers	customer_id (PK), gender, senior_citizen, partner, dependents	One row per customer: demographic profile (reviewed under NFR-09).
subscriptions	subscription_id (PK), customer_id (FK), contract_type, tenure_months, paperless_billing, payment_method, internet_service, phone_service, add-on service flags	Contract and service information.
billing	billing_id (PK), customer_id (FK), monthly_charges, total_charges	Billing summary (a true transaction history if available).
support_tickets (optional)	ticket_id (PK), customer_id (FK), opened_date, category, resolved_flag	Only if the dataset contains support information.
churn_labels	customer_id (PK, FK), churn_flag, observation_period	Target variable with its documented definition.
predictions (output)	customer_id (PK, FK), model_name, risk_score, risk_category, scored_at	Model output used by the dashboard.

Relationships: customers 1-to-many subscriptions; customers 1-to-1 churn_labels; customers 1-to-many billing and support_tickets; customers 1-to-many predictions (one per model).

Appendix B - Business Questions to Answer with SQL

#	Business question	Technique
Q1	How many customers do we have, and what is the overall churn rate?	COUNT, conditional aggregation
Q2	What is the churn rate for each contract type (month-to-month, one year, two year)?	GROUP BY
Q3	How does churn change across tenure bands (0-12, 13-24, 25-48, 49+ months)?	CASE WHEN, GROUP BY
Q4	Do customers with higher monthly charges churn more? (charge bands)	CASE WHEN, GROUP BY
Q5	Which internet service type has the highest churn?	JOIN, GROUP BY
Q6	Does payment method relate to churn?	JOIN, GROUP BY
Q7	How much monthly revenue was lost to churned customers, and what share of total revenue is that?	SUM, ratio
Q8	Do customers with more add-on services (security, backup, tech support) churn less?	Aggregation of service flags
Q9	Which combination of contract type and internet service has the highest churn (top 5 segments with at least 100 customers)?	GROUP BY, HAVING, ORDER BY, LIMIT
Q10	Rank customers within each contract type by monthly charges (window function) and show the churn rate of the top 10%.	Window functions (RANK / NTILE)
Q11	Data-quality check: which customer IDs are duplicated or have missing charges?	GROUP BY HAVING, IS NULL
Q12	If support data exists: do customers with multiple tickets churn more?	JOIN, aggregation

Appendix C - Suggested GitHub Repository Structure

```
customer-churn-project/
|-- README.md
|-- requirements.txt
```

```
|-- docs/
|   |-- Customer_Churn_Project_Documentation.pdf
|-- data/
|   |-- README.md          (source, license, data dictionary)
|-- sql/
|   |-- 01_schema.sql
|   |-- 02_load_data.sql
|   |-- 03_churn_analysis.sql
|-- notebooks/
|   |-- 01_data_cleaning_eda.ipynb
|   |-- 02_model_training_evaluation.ipynb
|-- src/
|   |-- preprocessing.py
|   |-- modeling.py
|-- outputs/
|   |-- model_comparison.csv
|   |-- scored_customers.csv
|-- dashboard/
|   |-- app.py
```

Do not commit large or restricted datasets unless the license and course rules allow it. Add a .gitignore for credentials and large files.

Appendix D - Glossary and Final Submission Checklist

Term	Meaning
Churn	A customer stopping use or payment of the service during the observation period.
Churn rate	Churned customers divided by total customers in the period.
Tenure	Number of months a customer has been subscribed.
Precision	Of the customers flagged as likely to churn, the share that actually churned.
Recall	Of all customers who actually churned, the share the model flagged.
F1-score	Harmonic mean of precision and recall.
ROC-AUC	Probability-like measure of how well the model ranks churners above non-churners across thresholds.
Confusion matrix	Table of true/false positives and negatives.
Lift	How many times better than random selection a model's top-risk group captures churners.
Data leakage	Using information in training that would not be available at prediction time, or letting test data influence preprocessing.
Class imbalance	One class (here, churners) is much less frequent than the other.

Final submission checklist (documentation, due Monday 19-10-2026)

- Project name chosen and written in the Drive project sheet.
- One GitHub repository created for the team (only one link) and written in the Drive sheet.
- Member names filled in on the cover page.
- All three parts (Planning & Management, Literature Review, Requirements Gathering) merged in one PDF.
- PDF uploaded to the repository (docs folder) and repository link opens for the instructor.
- Team Leader submits the 'Graduation Project' assignment on the LMS.
- Team confirms the final dataset and dashboard tool, and updates this document if implementation differs.
- References checked and cited consistently.